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Mailings Optimization

1. Problem Description

Center for Civic Engagement (CCE) is known for organizing different kinds of engaging events. As it is stated on their website, "*Civic engagement is when people are not indifferent to what is happening in their region, city or country*". Therefore, it is important for them to gather as many people as possible for the events organized by them. To do so, it is quite necessary for the contents of emails sent by them to be comprehensive and attractive to potential readers simultaneously. For the purpose of understanding better what features are the most influential on the attractiveness of an email we can apply principles from Machine Learning (ML) to find the golden mean of an email's contents. Thus, we decided to conduct a survey alongside CCE to get to know what people think of the emails sent by them.

2. Theoretical Models of the Problem

We decided that the best theoretical models for this project are K-Modes Clustering and Logistic Regression.
K-Modes Clustering: Standard clustering algorithms rely on Euclidean distance, however, the data from our survey isn't numerical, so we can't measure the distance between some values. K-Modes solves this by using dissimilarity based on matching categories. It minimizes the total number of mismatches between objects and cluster centers by finding the mode, instead of calculating a mean for each attribute in the cluster. Here's the math formula:

$$d(X, Y) = \sum_{i=1}^{m} \delta(x_i, y_i)$$

Where d is the distance between two categorical objects X and Y with m attributes. If the values match $\delta(x_i, y_i) = 0$, and 1 if they don't. A good example is that one cluster might be a student who prefers short subject lines and emojis in the text, while another might be a staff member who prefers links at the end with longer bodies.

Logistic Regression: In this project Logistic Regression determines what factors make an email attractive to potential readers. It's used for classification purposes, as it predicts whether someone belongs to the "Reader" and "Non-reader" group. It uses sigmoid function to map any input to a probability between 0 and 1. If the probability is ≥ 0.7, it's rounded up to 1 (Reader), and if it's < 0.7, it's rounded up to 0 (Non-reader). The threshold of 70% was chosen due to the fact that some factors could possibly cancel out each other – for instance, if a respondent likes "Opportunities", yet dislikes long body texts, the model spits out 0.5, which by default is rounded up to 1. Here's the math formula:

$$P(y = 1 | x) = 1 / (1 + e^{-(\beta_0 + \beta_1 x_1 + ... + \beta_n x_n)})$$

Where $P(y = 1 | x)$ is the probability that a respondent belongs to the first class (Reader) or not, β_0 is the y-intercept, β_n *is the learned coefficient for* x_n and $e^{-(...)}$ is the exponential function.

3. Description of the Raw Data and the Procedure of Collecting it

The dataset used in this project consists of 46 responses collected from Google Forms survey. Each row represents one respondent, and there are 20 columns including timestamp and so forth. In total, 38 students (82.6%), 5 professors (10.9%), and 3 staff members (6.5%) participated in the survey, and we are so thankful to them for doing so. Yet this makes the dataset being dominated by students, which is some sort of a limitation, since general preferences may not generalize to professors and staff.

The survey covers five thematic areas. First, reading behaviour: whether respondents read CCE emails (Q1), and what specific topics they read about (Q1.1). Second, subject line preferences: preferred word count (Q3), attractive keywords (Q2), and emoji appropriateness (Q4). Third, email body preferences: body word count (Q5), use of colored text (Q6), link placement (Q7), and delivery format (Q8). Fourth, engagement features: preferred reminder timing (Q9) and what a good email should have (Q10). Fifth, satisfaction: four Likert-scale statements (Q11a–e) rating deadlines visibility, formatting readability, engagement, opportunity visibility, and overall satisfaction. We decided to use only four Likert-scale statements, so that there won't be "Neutral" votes, which do not give us any info.

All 18 modelling columns are categorical with different variable types (binary, ordinal single-choice, 4-point likert scale, nominal single-choice and multiple-choice). Due to sparsity, Q1.1 and Q8.1 columns were dropped entirely, as they were shown only to specific readers, thus, 40 and 36 values respectively were missing for each of them. Q12 was also dropped, as it is an open-ended non-mandatory question, hence why there are 8 non-trivial responses, out of which four are completely non-informative ('.', 'Thanks', 'none' and 'not' ones specifically).

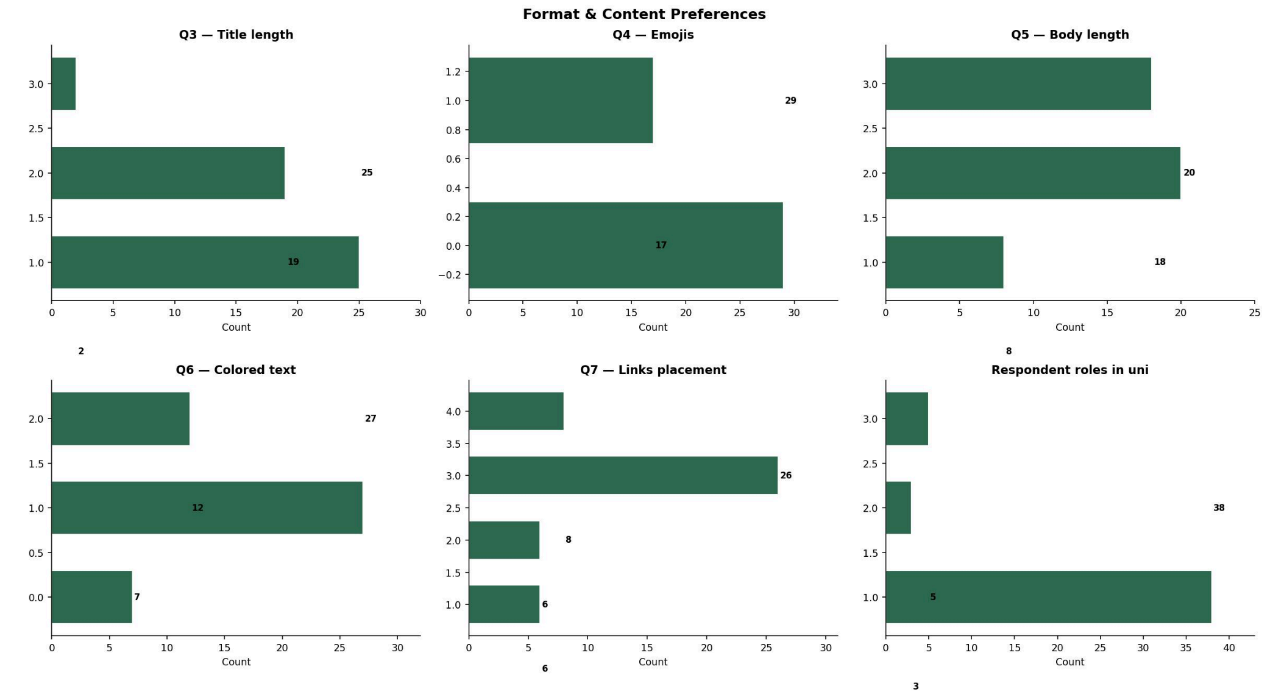
It took us a few weeks to gather the data. All of the respondents are related to AUCA, and most of them are students, because most professors and staff members didn't agree to participate in the survey, despite us asking them to do so.

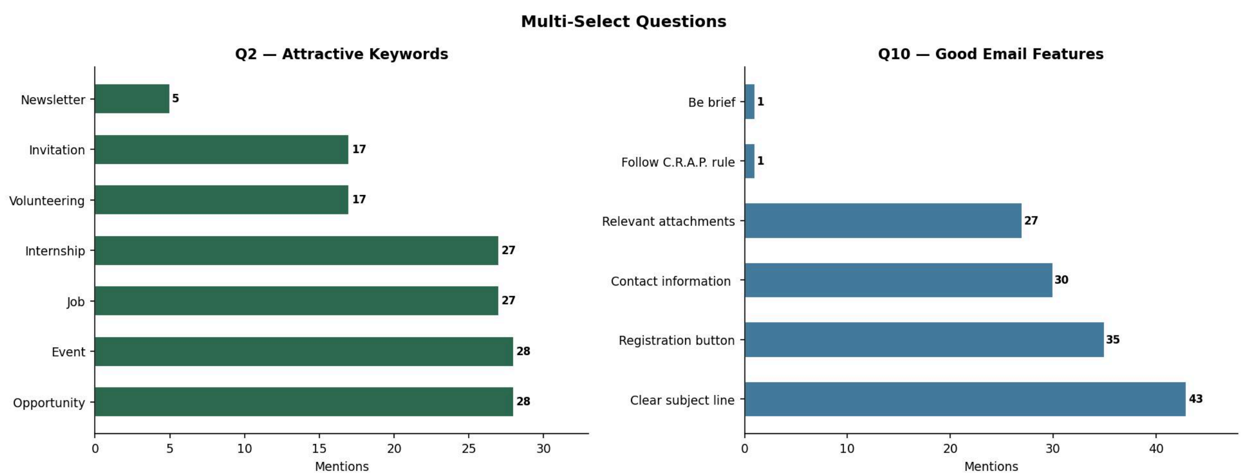
4. Data Cleaning

First of all, we renamed each column to a short key (i.e., 'Question 1' became 'q1'). Then we dropped irrelevant columns (Q1.1 and Q8.1 due to sparsity, and Q12, as the suggestions written there cannot be encoded for the chosen models, and also there was sparsity too). We also merged duplicate response values in the sixth question, as it contained two identical responses, which we merged into a single one not to treat them as separate classes during the encoding process. We also used whitespace normalization, so that invisible spacing won't treat two identical responses as different ones, and handled conditional nulls. Also, two long free-text responses from the second question were excluded during parsing, because they would've added noise to the data. Overall, after cleaning the data completely, the dataset retained 46 rows with 15 columns ready for encoding. The total number of remaining null values across all modelling columns was reduced to zero.

5. Data Visualization


CCE Emails Optimization — Analysis Dashboard
K-Modes Cost Curve
Cost
k
k = 3
Clusters (PCA)
PC1 (14.1%)
PC2 (9.9%)
C0 (n = 14)
C1 (n = 11)
C2 (n = 21)
Cluster Satisfaction
Dead
Form
Enga
Oppo
Sati
C0
C1
C2
3.29 3.07 2.64 2.43 2.64
3.55 3.09 2.55 2.09 2.55
3.43 3.29 2.90 2.24 3.00
Mean (1 – 4)
Feature Coefficients (reading behaviour)
↑ reading
↓ reading
engaging
satisfied
kw: Job
q3
rem: A day before the eve
kw: Opportunity
(posters, photos, and etc.)
rem: A few hours before t
deadlines
role
LR Coefficient
ROC — Logistic Regression
AUC = 1.00
TPR
FPR
Mann-Whitney A/B: C0 vs C1 (* p < 0.05)
Mean (1 - 4)
C0
C1
Q1 — Reading Frequency
All of them
Most of them
Sometimes
Only specific ones
5
14
18
6





These graphs show how often people actually read their emails. It can be seen here that the majority of people fall into the “Sometimes” category. It means that most students are not reading every email CCE sends, but rather visiting and not reading them. It is because they only open emails that look interesting, the subject line is the most important part. If the title does not attract them, then they will not read the rest. The Logistic Regression is used to figure out which parts of the title make people want to click.

The dashboard shows the “look” of the email. Majority of respondents want short titles (3 to 7 words) and short bodies (under 100 words). It also confirms that almost everyone answering the survey is a Student. Students are busy and usually check their email on their phones. If a title is too long, it gets cut off on the screen. If the body is too long, they won't finish it. This graph proves that "Short and Sweet" is the best rule. We use this data in our **K-Modes** math model to group people into "Personas" so we can send the right length of email to the right group.

We asked people which words catch their eye the most. The words **"Job"** and **"Internship"** were the big winners. People also really liked "Registration Buttons" and "Clear Subject Lines."

This is like a "Cheat Sheet" for the CCE. It shows that students care most about their future careers. If the CCE wants more people to open their emails, they should put the word "Job" right at the start of the title. We use these results to calculate a "Probability Score," which helps us predict if a new email will be a success or a failure based on the words used.

6. Brief Description of the Theory including Chosen Algorithms Justification

This project uses two machine learning methods: K-Modes Clustering and Logistic Regression. These algorithms were chosen because they fit the type of survey data we collected and help answer two different parts of the problem. K-Modes Clustering is suitable for categorical data. Since most of our survey answers are categories rather than continuous numbers, K-Modes is more appropriate than K-Means. Instead of using averages, it uses the most common category, or mode, inside each cluster. In our project, K-Modes helps group respondents with similar email preferences, such as title length, body length, emoji use, link placement, and email format. This is useful because different people may prefer different types of emails. The cost curve showed that three clusters was a reasonable choice for the final model. Logistic Regression was chosen because our main prediction task is binary classification. We wanted to predict whether a person is likely to be a reader or non-reader of CCE emails. In the code, respondents who selected anything except “None of them” in Question 1 were treated as readers. Logistic Regression is useful because it is simple, interpretable, and shows which factors increase or decrease the chance of reading emails. Together, these two methods help us understand both audience groups and reading behavior.

7. Solution of the Problem

To solve the problem, we first collected survey responses through Google Forms and prepared the data for analysis in Python. The dataset was cleaned by removing irrelevant columns, fixing repeated or inconsistent labels, and turning the answers into a format the models could use. Questions with too many missing values or open text answers that could not be encoded were dropped. Then, ordinal responses were converted into numbers, binary responses were changed into 0 and 1, and multi-select questions were transformed into separate indicator columns.

After cleaning, we created the target variable for Logistic Regression. If a respondent read at least some CCE emails, they were labeled as a reader. If they selected “None of them,” they were labeled as a non-reader. Then we applied K-Modes Clustering to find groups of respondents with similar preferences, and the final model used three clusters. Next, we trained the Logistic Regression model. Before training, the data was standardized and balanced with SMOTE so the model would not be too affected by class imbalance. The results showed that some factors were more strongly connected with reading behavior than others. In general, features related to engagement, satisfaction, opportunities, and job-related content were more likely to increase reading interest.

8. Analysis and Limitations of the Model

The analysis gave useful results, but it also has some clear limitations. One important finding is that respondents could be separated into three different preference groups, which shows that the audience is not fully the same. This means that one email format may not work equally well for everyone. The Logistic Regression model also helped us see which features were linked to stronger reading behavior. At the same time, the project has limitations. The biggest one is the small dataset, since we only had 46 responses. This makes the model less reliable and makes it harder to generalize the results to a larger population. Also, most respondents were students, so the findings mainly reflect student preferences more than staff or professors. Another limitation is that the survey shows what people say they prefer, not what they actually do when they receive an email. Real open rates and click data might show a slightly different picture. Also, reading behavior was simplified into only two groups, even though in real life people engage with emails at different levels. Because of this, the model should be seen as a useful guide, not a perfect final answer.

9. Possible Predictions Based on the Above Solution

Based on the results, we can predict that CCE emails are more likely to be read when they have clear subject lines and include keywords such as opportunity, event, job, and internship. Respondents also seemed to value practical features like a registration button, contact information, and relevant attachments. The results also suggest that people may respond better to emails that are not too long, easy to follow, and written in a more engaging way. Since the clustering model showed different preference groups, it is also reasonable to predict that CCE could get better results by adjusting email style for different audiences instead of using exactly the same format for everyone.

10. Conclusion and Practical Recommendations

In conclusion, this project shows that machine learning can be used in a practical way to improve CCE email communication. K-Modes Clustering helped us understand different groups of respondents, while Logistic Regression helped us identify which email features are more closely linked to reading behavior. Together, these methods gave us useful insight into how email content and structure affect engagement. The main practical recommendation is that CCE should focus on making emails clear, relevant, and easy to act on. Subject lines should be direct and include attractive words like opportunities, events, jobs, or internships. The body of the email should be organized and not too long, and important actions such as registration should be easy to find. It may also help to test slightly different formats for different groups of readers.
Even though the dataset is small, the project still gives useful guidance. It shows that better email design can improve communication and make CCE messages more likely to be noticed and read.

11. Impact of Each Group Member

All of us contributed a lot to this project, thus, it's quite hard to differentiate between any of us. We did all the things together, and, probably, share the same amount of influence on the project.

12. List of References

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